MIT 18.05 Introduction to Probability and Statistics
Cheat sheet 2

Yann Fonkoue

1 Introduction to Discrete Random Variables

A **discrete random variable** X is a **function** from a **sample space** Ω to the real numbers $\mathbb{R}$ that takes a **discrete set of values**. Its value is **"random"** because it depends on the outcome of a **random experiment**.

1.1 Probability Functions

- **Probability Mass Function (pmf):** Defined as $p(a) = P(X = a)$. It satisfies $0 \leq p(a) \leq 1$ and the **sum of probabilities over all possible values is 1**.
- **Cumulative Distribution Function (cdf):** Defined as $F(a) = P(X \leq a)$.
- **Properties of the cdf:** F is **non-decreasing**, ranges from **0 to 1**, and satisfies

$$\lim_{a\to\infty} F(a) = 1, \qquad \lim_{a\to-\infty} F(a) = 0.$$

2 Specific Discrete Distributions

The course focuses on four **fundamental discrete distributions**:

1. **Bernoulli(p):** Models a **single trial** with **success** (1) or **failure** (0). $P(X = 1) = p$.
2. **Binomial(n, p):** Models the **number of successes** in n **independent Bernoulli trials**.
3. **Geometric(p):** Models the **number of failures before the first success**.
4. **Uniform(N):** Models N **equally likely outcomes**, each with probability $1/N$.

3 Expected Value (Mean)

The **expected value** $E[X]$, or **mean** μ, is a measure of **central tendency** representing the **"center of mass"** of the distribution.

3.1 Definitions and Properties

- **Formula:** $E[X] = \sum p(x_j)x_j$.
- **Linearity:** $E[X + Y] = E[X] + E[Y]$ (for any X, Y) and $E[aX + b] = aE[X] + b$.
- **Functions of a Variable:** $E[h(X)] = \sum h(x_j)p(x_j)$. Note that $E[h(X)] \neq h(E[X])$ in general.

4 Variance and Standard Deviation

Variance $Var(X)$ measures the **"spread"** of the probability mass around the mean. The **standard deviation** is $\sigma = \sqrt{Var(X)}$.

4.1 Key Formulas and Properties

- **Definition:** $Var(X) = E[(X-\mu)^2] = \sum p(x_i)(x_i-\mu)^2$.
- **Alternative Formula:** $Var(X) = E[X^2] - (E[X])^2$.
- **Scaling and Shifting:** $Var(aX+b) = a^2 Var(X)$.
- **Sum of Independent Variables:** If X and Y are **independent**, $Var(X+Y) = Var(X) + Var(Y)$.

5 Summary Table of Distributions

Distribution	**pmf** $p(k)$	**Mean** $E[X]$	**Variance** $Var(X)$
Bernoulli(p)	$p^k(1-p)^{1-k}$	p	$p(1-p)$
Binomial(n,p)	$\binom{n}{k}p^k(1-p)^{n-k}$	np	$np(1-p)$
Geometric(p)	$(1-p)^k p$	$\frac{1-p}{p}$	$\frac{1-p}{p^2}$
Uniform(n)	$1/n$	$\frac{n+1}{2}$	$\frac{n^2-1}{12}$